BCS ICT FEST 2026 • PROJECT SHOWCASE

UN SDG 9: INFRASTRUCTURE • SDG 8: ECONOMIC GROWTH

MDEFENDER PRO: Autonomous Cloud WAF & Cyber Resilience Ecosystem

An Enterprise-Grade, Self-Healing Web Application Firewall & AI Threat Defense Platform

Live Platform: https://mdefender-pro.crafix.tech
Team Name: Team CyberShield
Category: Cybersecurity & Software Innovation
Target Infrastructure: Global Web & WordPress Fleet Defense

1. EXECUTIVE SUMMARY & PROBLEM STATEMENT

The rapid expansion of the digital economy has made web platforms critical national infrastructure, yet open web ecosystems remain intensely vulnerable. Over **43% of all global websites**, powering SMEs, universities, healthcare portals, and public services, rely on open CMS architectures like WordPress. These systems face relentless attacks—averaging over **90,000 automated cyber-attacks every minute**, ranging from SQL Injection (SQLi) and Remote Code Execution (RCE) to credential-stuffing botnets and Layer-7 DDoS extortion. Current enterprise security solutions (such as Cloudflare Enterprise or Imperva) cost thousands of dollars per month, lock users into opaque proprietary networks, and add significant latency to origin servers.

MDefender Pro eliminates this disparity by delivering a high-performance, cost-effective, and autonomous Web Application Firewall (WAF) ecosystem. Engineered with an asynchronous Python FastAPI cloud backend and a zero-dependency PHP client-side service container, MDefender Pro provides edge-level packet filtering, sub-millisecond AI attack classification, automated malware checksum auditing, and a real-time Core Web Vitals speed observatory at near-zero marginal cost.

2. DECOUPLED 3-TIER SAAS CONTROL-AND-COMMAND ARCHITECTURE

MDefender Pro employs a high-throughput decoupled architecture that completely isolates edge request processing from central telemetry storage, preventing database bottlenecks even under severe attack floods:

Tier 1: Client Edge Agent

- **Zero-Dependency PHP Container:** Executes before WordPress core hooks to block attacks with zero bootstrap latency.
- **Circuit Breaker Isolation:** Sandboxes all 17 engines with `try...catch(\Throwable)` to guarantee 100% site availability.
- **Zero API Blocking:** Evaluates WAF rules locally without waiting for external cloud HTTP calls.

Tier 2: Cloud Control Plane

- **FastAPI Async Python Server:** Powers centralized fleet management, token auth & event streaming.
- **Zero-Bloat MongoDB:** Stores only fleet metadata and configuration (all logs stay distributed on client nodes).
- **Synthetic Uptime & SSL:** Automated Let's Encrypt certificates and edge health probing.

Tier 3: Telemetry Observatory

- **Modern React 18 (Vite):** Real-time management console and attack audit dashboard.
- **Fleet Speed Observatory:** Real-time Core Web Vitals (Mobile 4G vs. Desktop).
- **1-Click Security Hardening:** Instant activation of login hide, 2FA, & header policies.

3. MASSIVE 14-MILLION THREAT INTELLIGENCE & AI/ML PIPELINE

A key technical breakthrough of MDefender Pro is its **13.91M+ threat intelligence matrix**. Instead of burdening relational databases with millions of rows, data is pre-compiled into ultra-efficient indexed binaries that execute in micro-seconds:

Intelligence Matrix	Dataset Volume	Engine Architecture	Runtime Performance & Real-World Benefit
WordPress Malware & Core Checksums	**5,240,000+ Records**	Pre-hashed SQLite B-Tree Index (`wp_checksums.db`)	0.01ms file integrity verification; pinpoints backdoors, web-shells & code tampering instantly.
Threat Intelligence & CVE Signatures	**1,140,000+ Records**	In-Memory Radix Trie & Indexed SQLite Signatures	Instant IP reputation and zero-day exploit blocking before requests reach web application core.
WAF Attack Request Vectors	**7,520,000+ Vectors**	Machine Learning SGD Model (`waf_model.pkl`, 129 KB)	Real-time semantic request evaluation; sub-millisecond AI classification with zero cloud bloat.

4. INTEGRATED SECURITY MODULES & ENTERPRISE FEATURES

1. Next-Gen Web Application Firewall (WAF)
Deep inspection of GET, POST, URI, and HTTP headers against SQLi, XSS, Remote Code Execution (RCE), and Path Traversal with custom regex and AI heuristics.

2. Proof-of-Work (PoW) Anti-DDoS Challenge
Under-attack protection running cryptographic in-browser SHA-256 compute challenges. Nullifies 100% of Layer-7 bot floods without frustrating visual CAPTCHAs.

3. Stealth Admin Gateway & CIDR Whitelist
Obfuscates default `/wp-login.php` with cryptographic secret tokens and restricts admin dashboard access strictly to authorized IP subnets via 403 barriers.

4. Native RFC-6238 2FA & Security Hardening
Enforces authenticator app TOTP (Google/Microsoft Authenticator), file upload code-execution bans, directory lockdown, and security headers (HSTS, CSP).

5. Core Web Vitals Speed Observatory
Dual simulated Mobile (4G) and Desktop speed auditor reporting authentic LCP, FCP, CLS, TTFB, and actionable asset optimization recommendations with zero fake values.

6. Cloud Malware Scanner & Quarantine
High-speed signature and AST-heuristic scanner comparing client filesystem against 5.24M official WordPress checksums to detect and isolate zero-day web-shells.

5. ALIGNMENT WITH UN SUSTAINABLE DEVELOPMENT GOALS (SDGS)

SDG 9: Industry & Infrastructure
Fortifies national and global digital infrastructure against systemic cyber warfare, automated botnets, and downtime, fostering a resilient digital economy.

SDG 8: Decent Work & Growth
Shields online businesses, SMEs, and digital entrepreneurs from crippling revenue loss, customer data breaches, defacement, and extortionate ransom demands.

SDG 16: Strong Institutions
Protects academic institutions, public healthcare repositories, and civic portals from malicious data manipulation, ensuring digital integrity and public trust.

6. EMPIRICAL VERIFICATION & PRODUCTION PERFORMANCE BENCHMARKS

MDefender Pro is fully deployed and accessible live at **https://mdefender-pro.crafix.tech**. Rigorous testing across multi-domain production environments confirms top-tier reliability:

< 1.05 ms	100%	13.91M+	0 ms
WAF INSPECTION OVERHEAD	L7 FLOOD MITIGATION (POW)	ACTIVE THREAT RECORDS	BLOCK PAGE PAINT SHIFT

7. FUTURE INNOVATION ROADMAP & COMMERCIALIZATION

The development roadmap for MDefender Pro advances toward an autonomous global cybersecurity network:

- **Distributed Anycast Edge Mesh:** Expanding edge reverse-proxy nodes across regional datacenters to filter traffic closer to users.
- **Automated AI Virtual Patching:** Ingesting newly disclosed CVEs into real-time WAF regex directives within 15 minutes of zero-day discovery.
- **Federated Threat Intelligence Network:** Anonymized cross-fleet attack telemetry sharing to immunize all connected sites against emerging botnets.
- **Commercial SaaS Rollout:** Providing affordable subscription tiers for Bangladeshi software firms, educational universities, and global enterprises.

8. Conclusion

MDefender Pro transitions cybersecurity from a reactive, costly chore into an accessible, autonomous, and high-performance utility. Combining 14 million threat vectors, sub-millisecond execution, and comprehensive fleet visibility, MDefender Pro demonstrates that student innovation can produce enterprise-grade cyber sovereignty that secures the digital future.